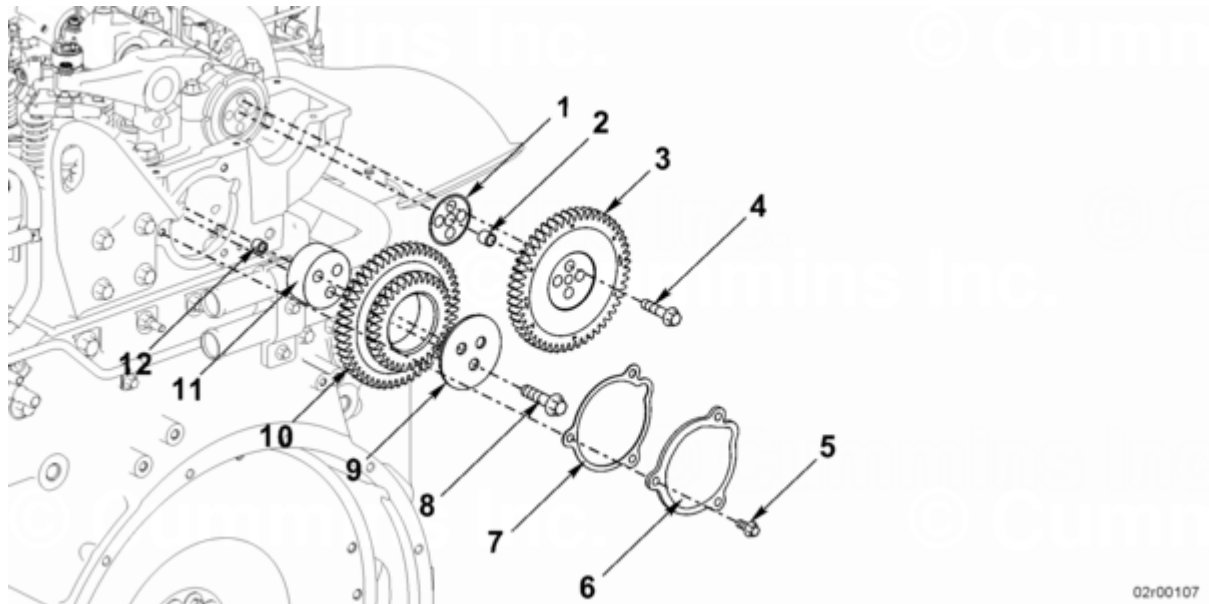


Trainings ▾

Exploded View



Overhead Camshaft Gear Exploded View

1. Friction shim - quantity 1
2. Ring dowel - quantity 1
3. Overhead camshaft gear - quantity 1
4. Capscrew - quantity 4
5. Capscrew - quantity 3
6. Cover plate - quantity 1
7. Cover plate gasket - quantity 1
8. Capscrew - quantity 3
9. Gear retainer - quantity 1
10. Compound camshaft idler gear - quantity 1
11. Idler shaft - quantity 1
12. Ring dowel - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- Engine barring tool, Part Number 4919092
- Engine Timing Kit, Part Number 5572725

Additional Service Items

- No additional service items required.

General Information

If the camshaft gear is separated from camshaft, verify a new friction shim is used.

The four capscrew holes on camshaft gear and three capscrew holes on compound camshaft idler gear shaft are **not** symmetrically arranged.

There are two layers camshaft idle gear on this engine. The compound camshaft idler gear is installed on cylinder head, and another is installed on cylinder block (middle idle gear). Refer to Procedure 001-115 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-115.html>)

Preparatory Steps

WARNING

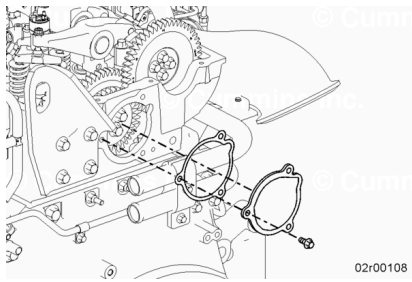
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/1016/1016-003-011-tr.html>)

Remove

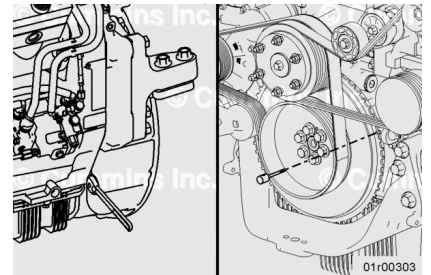
Remove the cover plate.

Discard the cover plate gasket.



Use the engine barring tool, Part Number 4919092, to rotate the engine **clockwise** when viewed from the crankshaft pulley to Top Dead Center (TDC).

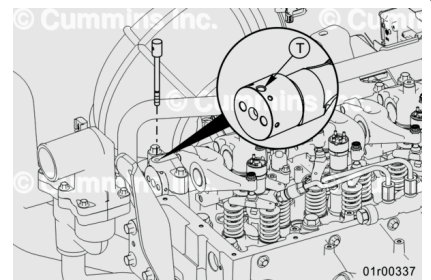
Use the timing pin, Part Number 5572723, included in Engine Timing Kit, Part Number 5572725, to lock the engine by pushing it through the designated holes on the vibration damper and the lubricating pump housing.



⚠ CAUTION ⚠

Engine damage can occur if the static timing of the engine is not correct. The crankshaft and camshaft must be pinned before removing the camshaft gear.

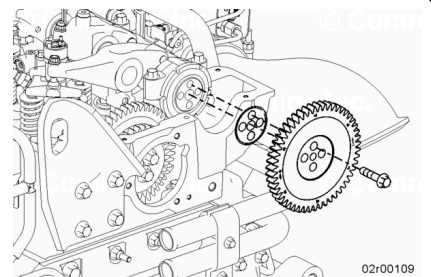
Insert the cam timing pin, Part Number 5572724, included in Engine Timing Kit, Part Number 5572725, into the bigger hole found on the front of the camshaft. The camshaft will be locked at TDC for number 1 cylinder.



Remove the four bolts from the camshaft gear.

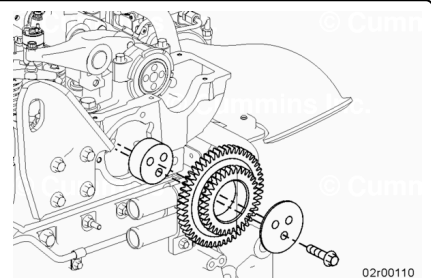
Remove the camshaft gear.

Discard the friction shim.



Remove the three bolts from the compound camshaft idler gear.

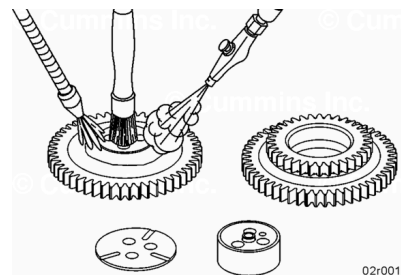
Remove the compound camshaft idler gear, shaft, and gear retainer.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

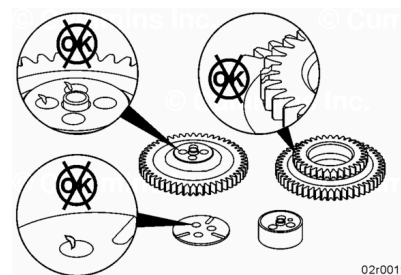
Clean the overhead camshaft gear, compound camshaft idler gear, shaft, and gear retainer with solvent.

Dry with compressed air.

Inspect the gear, shaft, and gear retainer for cracks or excessive wear.

Check for missing or bent teeth.

Replace if any damage is found.



Install

Install the compound camshaft idler gear to shaft.

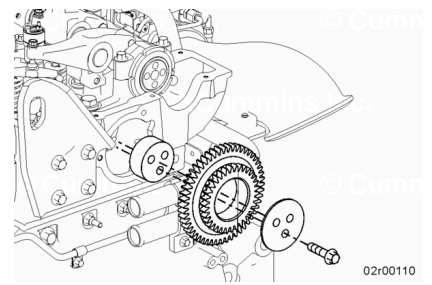
Install the gear with shaft onto block.

Verify ring dowel on shaft is engaged with hole in block.

Install gear retainer with capscrews.

Tighten all three cam idler gear capscrews.

Torque Value: 120 n·m [89 ft-lb]



Install the overhead camshaft gear with new friction shim and tighten the four bolts.

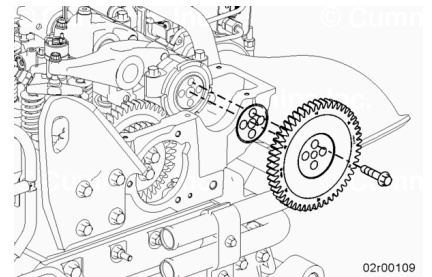
Verify ring dowel is on gear.

Torque Value:

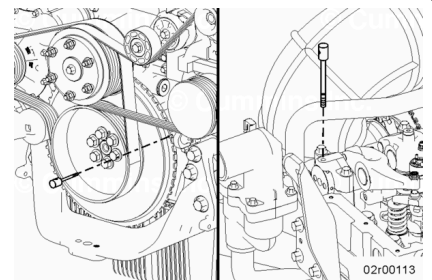
Spur gear 65 n·m [48 ft-lb]

Torque Value:

Helical gear 76 n·m [56 ft-lb]



Remove the timing pins.

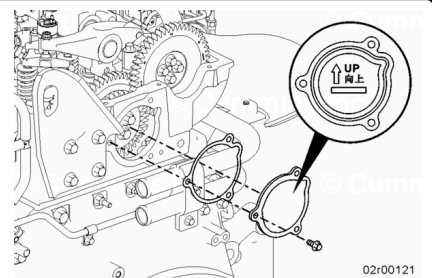


Install the cover plate with new gasket.

Torque Value: 23 n·m [204 in-lb]

Note : Be caution of the tang location on the cover plate and gasket.

Note : Put the arrow direction vertically and toward up.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/1016/1016-003-011-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.

Last Modified: 19-Nov-2024
